KOROGNAISMO

Korognai SMO — Reference Implementation

Full Documentation — RAN, and the Cross-Domain SMO Concept (RAN + Core + Transport)

Every tab, every term, every scenario explained — across both experiences shipped in this build.

Synthetic demo, no live network or vendor affiliation.

This document explains, in full detail, every tab, metric, term, and simulated scenario inside the Korognai SMO reference demo. It is written for two audiences at once: someone with deep telecom background who wants to verify the technical grounding, and someone with no telecom background who wants to understand what they're looking at and why it matters.

What this demo is: a single, self-contained HTML file that runs entirely in a web browser, with no backend server and no live network behind it. Every number, alarm, and event on screen is generated live, in-browser, by JavaScript — a working simulation of how a real SMO platform behaves, not a recording or a mockup.

What this demo is not: it is not connected to any real radio network, it does not use any vendor's proprietary software or confidential material, and no named company, product, or trademark of any commercial SMO vendor appears in it. Only generic, industry-standard terminology drawn from public O-RAN Alliance, 3GPP, TM Forum, ETSI, ITU, ONF, GSMA, NIST, and CIS material is used.

*This build ships **two linked experiences**. Sections 3–19 document the original, RAN-focused site. Section 20 documents the **Cross-Domain SMO concept**, reachable from the main site's header via a pulsing "Want to see into the future?" card: it takes the identical closed-loop philosophy — rApps on the Non-RT RIC — and extends it across **Core** (cApps) and **Transport** (tApps) as well as RAN, adding cross-domain topology, a merged three-domain closed loop, intent-based assurance, cross-domain conflict detection (including conflicts between domains, not just within one), and a Future & Standards Radar.*

Table of Contents

1. Purpose & How to Use This Document

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The document walks through the original site in the same order its tabs appear, left to right (Sections 3–18), closes that portion with Design Principles (Section 19), then covers the Cross-Domain SMO concept end to end (Section 20), and finishes with an expanded reference Appendix (Section 21). Section 2 defines every acronym and concept used anywhere in *either* experience, so later sections can reference them without re-explaining each time.

2. Core Concepts & Glossary

This section covers terms from *both* experiences — the original RAN vocabulary (2.1–2.10) plus terms specific to Core, Transport, and cross-domain/standards-body concepts (2.11–2.14).

2.1 The physical radio network, in plain terms

Term	Full name	Plain-English meaning
RU	Radio Unit	The physical hardware at the antenna. Converts digital signals into actual radio waves and back. Stays at the tower.
DU	Distributed Unit	Handles time-critical, millisecond-scale processing, usually located physically close to the tower.
CU	Centralized Unit	Handles less time-critical processing; can serve many towers from one place. Split into CU-CP (control) and CU-UP (user data).
Core	Core Network	The wired part of the network the radio side connects into — routes traffic to the internet or other networks.

Segment	Connects	Timing sensitivity
Fronthaul	RU ↔ DU	Extremely high timing sensitivity — microseconds
Midhaul	DU ↔ CU	High timing sensitivity — single-digit milliseconds
Backhaul	CU ↔ Core	Moderate timing sensitivity

2.2 The management & intelligence layers

Term	Full name	Plain-English meaning
SMO	Service Management & Orchestration	The top-level management "brain" overseeing the whole radio network from above.
Non-RT RIC	Non-Real-Time RAN Intelligent Controller	The slower (>1s, often minutes+) intelligence layer inside the SMO. Hosts rApps.
Near-RT RIC	Near-Real-Time RAN Intelligent Controller	A separate platform, closer to the radio, deciding on a 10ms–1s timescale. Hosts xApps.
rApp	RIC app	An automated behavior hosted on the Non-RT RIC — e.g. energy saving, mobility tuning.
xApp	RIC app	An automated real-time behavior hosted on the Near-RT RIC — e.g. dynamic spectrum sharing.

2.3 Interfaces — how the layers talk to each other

O-RAN Alliance defines a set of open, standardized interfaces so equipment from different vendors can interoperate. This is the single most important reference table for how any part of the RAN experience connects to any other.

Interface	Connects	Purpose
O-FH	RU ↔ DU	Standardized Open Fronthaul (functional split 7.2x) — enables multi-vendor RU/DU interoperability.
F1	DU ↔ CU	Midhaul interface, split into F1-C (control) and F1-U (user plane).
E1	CU-CP ↔ CU-UP	Separates control-plane and user-plane processing within the CU.

Interface	Connects	Purpose
E2	Near-RT RIC ↔ RU/DU/CU	Real-time RAN data up, real-time control down. The xApp control channel.
A1	Non-RT RIC (SMO) ↔ Near-RT RIC	AI/ML-informed policy guidance and enrichment data, passed down.
O1	SMO ↔ all managed RAN elements	FCAPS — fault, configuration, accounting, performance, security.
O2	SMO ↔ O-Cloud infrastructure	Cloud infrastructure and workload orchestration/lifecycle.
R1	rApps ↔ Non-RT RIC	How an rApp registers, declares data needs (SME/DME), and receives that data.

2.4 FCAPS — the five pillars of network monitoring

Letter	Stands for	Plain-English meaning
F	Fault Management	Tracking things that have gone wrong (alarms) so they can be acknowledged, cancelled, and fixed.
C	Configuration Management	A record of every setting on the network and every change made to it.
A	Accounting Management	Tracking usage and resource consumption (often for billing or capacity planning).
P	Performance Management	Tracking how well the network is actually running — speed, reliability, capacity.
S	Security Management	Access control, audit trails, and protecting the network from misuse.

2.5 Radio, performance & 5G traffic-type terms

PRB	Physical Resource Block — the basic unit of radio capacity a cell tower allocates. Utilization = % currently in use.
Handover (HO)	A phone switching from one tower's coverage to another's without dropping the call.
RRC	Radio Resource Control — the signaling protocol establishing a device's initial radio-level connection.
RSRP	Reference Signal Received Power — how strong a phone's signal from a given tower actually is.
UE	User Equipment — the device connecting to the network.
eMBB	Enhanced Mobile Broadband — the 5G use case optimized for raw speed and capacity.
URLLC	Ultra-Reliable Low-Latency Communication — optimized for guaranteed, near-instant delivery.
mMTC	Massive Machine-Type Communication — huge numbers of low-power sensors/devices at once.
GBR / PDB / PER	Guaranteed Bit Rate / Packet Delay Budget / Packet Error Rate.
CA	Carrier Aggregation — combining multiple frequency channels into one connection.
DL / UL	Downlink (network→device) and Uplink (device→network).

Network Availability	% of time the network is up and reachable — capped at 100%.

2.6 Automation concepts

Closed loop	Observe → decide → act → verify, fully automated, no human step in between.
SON	Self-Organizing Networks — the established term for this kind of automated optimization; rApps formalize SON under open interfaces.
Conflict management	What happens when two automated behaviors try to change the same thing in different directions at the same time (Section 11; Section 20.4 for the cross-domain version).
Intent-based management	Telling the network what outcome you want rather than how to achieve it.
Agentic AI	AI that can reason, plan, and take multi-step action toward a goal. The demo's Copilot narration is a deliberately modest example: it explains decisions in plain language.

2.7 Network slicing terms

Network slice	A virtual, separate network carved from one shared physical network, with its own guarantees.
NSSI	Network Slice Subnet Instance — the piece of a slice living in a given domain (RAN, core, or transport).
S-NSSAI	The identifier a network uses internally to tell which slice traffic belongs to.
5QI	A standardized code telling the network what QoS treatment a type of traffic needs.
SLA	Service Level Agreement — a measurable, contractual promise about network quality.

2.8 Cloud & infrastructure terms

O-Cloud	The O-RAN term for cloud infrastructure hosting virtualized network functions.
CNF	Cloud-native Network Function — network software packaged to run in containers.
vDU / vCU-CP / vCU-UP	Virtualized DU/CU running as software on standard servers.
NF Lifecycle (LCM)	Instantiate, heal, scale, upgrade, terminate.
O2-DMS	O2 Deployment Management Service — how the SMO tells cloud infrastructure what to run.
PnP	Plug-and-Play — new hardware announces itself and is configured automatically.
DC	Data Center — physical/regional facility hosting cloud-native network functions.

2.9 Security & trust terms

Zero-trust (NIST SP 800-207)	Nothing trusted automatically — every request checked, every time.
CIS Benchmarks	An independent, published checklist of secure configuration settings.
SBOM	Software Bill of Materials — an itemized list of every component inside a piece of software.

Supply-chain integrity	Proof the software running today is exactly what was built and approved.

2.10 Business & commercial terms

TCO	Total Cost of Ownership — full lifecycle cost, not just sticker price.
OPEX	Operating Expenses — the ongoing cost of running something.
OSS / BSS	Operations Support System / Business Support System — the operator's own systems north of the SMO.
SME / DME	Service Management & Exposure / Data Management & Exposure — the R1 services an rApp uses.
MTTR	Mean Time to Resolve.

2.11 The 5G Core (Core domain)

Introduced by the Cross-Domain concept (Section 20). The core handles registration, session management, policy and routing once traffic leaves the radio.

SBA / SBI	Service-Based Architecture / Service-Based Interface — every core function exposes an HTTP/2 API and discovers others via a registry, instead of fixed point-to-point links.
AMF	Access & Mobility Management Function — registration, connection and mobility management; terminates NAS signaling.
SMF	Session Management Function — PDU session establishment, modification, release; selects and controls the UPF.
UPF	User Plane Function — the only core function actually in the path of user data itself.
PCF	Policy Control Function — decides QoS and charging rules.
UDM	Unified Data Management — subscriber profile, credentials, subscription data.
NRF	Network Repository Function — service discovery registry.
NSSF	Network Slice Selection Function — picks which slice instance serves a device/session.
NWDAF	Network Data Analytics Function — the core's standardized analytics engine, roughly the Non-RT RIC's counterpart inside 5GC. Feeds cApps.
N2 / N3 / N4	Core reference points: N2 (RAN↔AMF signaling), N3 (RAN↔UPF user-plane), N4/PFCP (SMF↔UPF forwarding control).
SEPP / N32	Security Edge Protection Proxy — trust boundary for inter-operator roaming signaling, via the N32 reference point.
cApp	The Core-domain equivalent of an rApp — subscribes to NWDAF/SBI telemetry, closes the loop via core network function provisioning.

2.12 Transport (fibre, microwave & optical network)

T-SDN	Transport Software-Defined Networking — the controller programming the underlying optical/IP transport network, exposed northbound as paths rather than box-by-box CLI.
TAPI	Transport API — an ONF-defined open northbound interface for requesting/managing transport paths; the transport equivalent of O1/O2.
DWDM	Dense Wavelength Division Multiplexing — packs many optical signals onto one fibre using different wavelengths.
OSNR	Optical Signal-to-Noise Ratio — headroom above the minimum needed for error-free transmission.
PTP / SyncE	Precision Time Protocol / Synchronous Ethernet — distribute the precise timing split-RAN and 5G accuracy depend on.
tApp	The Transport-domain equivalent of an rApp — subscribes to TAPI telemetry and PTP/SyncE status, closes the loop via the T-SDN controller.
ONF	Open Networking Foundation — the standards body behind TAPI.

2.13 Cross-domain, standards bodies & the future-looking material

Non-RT RIC Federation	Emerging O-RAN work letting multiple Non-RT RICs — across vendors or regions — share rApp/analytics intelligence.

TM Forum / ETSI ZSM	TM Forum: Open APIs and the Autonomous Networks (L0–L5) maturity model. ETSI ZSM: the Zero-touch network & Service Management closed-loop reference architecture.
GSMA / CAMARA	GSMA's Open Gateway initiative and its CAMARA-defined developer APIs, exposing network capabilities to third-party apps.
IMT-2030 / ITU-R	The ITU-R's 6G framework recommendation and the UN standards body that issued it.
PQC	Post-Quantum Cryptography — encryption designed to resist future quantum-capable attacks; an emerging O-RAN WG11 scope item.
Network Digital Twin	A live, simulateable model of the network used to test a closed-loop action before it touches production — an ITU-T/operator-led direction, not yet in this build.

2.14 Intent-based management

Demonstrated on the Intent-Based Assurance tab (Section 20.5) using 3GPP TS 28.312's model:

Intent	A plain-language statement of the outcome an operator wants (e.g. "guarantee low latency for factory robotics") — 3GPP TS 28.312's top-level object, and TM Forum's IG1230/TMF921 Open API resource.
Expectation	A measurable sub-goal SMO derives from an Intent (e.g. "latency ≤ 5ms one-way") — the bridge between a plain-language goal and something a system can actually check.
Target	The concrete, per-domain numeric threshold an Expectation becomes — e.g. RAN handover failure rate ≤ 2.0%, Core UPF load ≤ 70%, Transport fronthaul latency ≤ 90µs. Targets are what compliance is actually checked against.
TMF921	TM Forum's published Intent Management API standard (the successor to draft IG1230) — defines intent as a resource an operator's systems can request, track, and query the fulfillment status of.

3. Overview Tab

The landing page, written for a non-technical reader — a VP or operations executive should understand the platform within about thirty seconds.

3.1 "In plain terms" callout

A single paragraph frames the whole platform as "the control room that runs a modern mobile network," deliberately avoiding all acronyms.

3.2 Executive KPI row

KPI	What it measures	Where the number comes from
Issues Resolved Automatically	How many times an rApp fixed something today with no human paged	A live counter incremented on every real rApp action across all six rApps
Est. Annual OPEX Saved	Illustrative savings estimate	A fixed illustrative figure — not a live measurement
Avg. Time to Detect & Resolve	How quickly an issue is fixed	A fixed illustrative figure, framed against a manual-process baseline

3.3 Operational KPI row

Active Alarms and Network Availability link to FCAPS Monitor; rApps in Closed Loop links to Closed Loop — the Overview doubles as a navigation hub.

3.4 The cluster grid

Eight clusters spanning the platform's full functional scope, each with a business-value line before the technical description, each clickable through to its tab.

4. RAN Architecture Tab

Answers, visually: how do SMO, the Non-RT RIC, and the Near-RT RIC divide responsibility, and what connects them?

4.1 The diagram

Three layers stacked top to bottom: SMO (Non-RT RIC/rApps) connecting down via O1/O2/A1; Near-RT RIC (xApps) via E2; the radio chain (RU→O-FH→DU→F1→CU-CP/CU-UP→backhaul→Core).

4.2 The hover-highlight interaction

Every interface label — in the diagram or the reference table — is wired to a shared highlight function; hovering either highlights both simultaneously, with centralized clear-on-hover-out logic so no highlight can get stuck.

4.3 Interface reference table

Identical to the table in Section 2.3.

5. FCAPS Monitor Tab

Core network monitoring discipline. Four scrollable panels — Active Alarms, Live KPIs, Configuration Management, Trace Sessions — share one consistent box size.

5.1 Fault Management — active alarms

100 alarms, each with severity, source, description, age, and independent Acknowledge / Cancel actions.

Critical	Something has completely stopped working — e.g. a site outage
Major	Significant degradation needing attention soon
Minor	A condition worth tracking but not yet service-impacting
Warning	An early trend worth watching

5.2 Performance Management — live KPIs

Ten KPIs on a bounded random walk (Section 5.2.1); hovering reveals a synthetic 24h trend plus min/max.

KPI	Bounds	Why it matters
RAN Availability	99.5%–100%	Cannot exceed 100% by definition
Call Setup Success Rate	95%–100%	Healthy but not impossibly perfect
Avg. DL Throughput / cell	120–260 Mbps	Realistic modern 5G cell range
Drop Call Rate	0.05%–2.5%	Low but never exactly zero
Avg. UL Throughput / cell	15–60 Mbps	Typically a fraction of downlink
PRB Utilization (avg)	40%–95%	How busy radio capacity is on average
Handover Success Rate	90%–100%	Cleanliness of inter-cell mobility
RRC Setup Success Rate	95%–100%	Initial radio connection reliability
Packet Error Rate (URLLC)	0.001%–0.5%	Near-zero-tolerance metric

KPI	Bounds	Why it matters
Avg. Latency (PDB)	2–15 ms	Measured against Packet Delay Budget

5.2.1 Why a bounded random walk

Each KPI takes a small random step from its current value, clamped to a realistic range — smooth, lifelike drift without impossible readings.

5.3 Configuration Management — recent changes

100 entries: timestamp, entity, parameter, old/new value, and who or what made the change — including specific rApps as change sources.

5.4 Trace sessions

Start a trace against a random cell; hover a completed session for a synthetic snapshot (RRC setup time, DL/UL throughput, PER, handovers, duration).

5.5 Topology & inventory summary

Rollup counts: sites, RAN network functions, distinct vendors integrated.

6. Closed Loop Tab & Scenario Reasoning

The centerpiece: the Non-RT RIC detecting and fixing problems automatically, with real O-RAN SON-style algorithms behind all six rApps.

6.1 The cell ring

Twelve cells across three simulated vendors around a Non-RT RIC hub; color = state (green/amber/red/cyan-remediating); hover for ID, vendor, state, live metrics, last action. The cyan indicator persists a few seconds so fast fixes stay visible.

6.2 Sample vs. fleet scale

A toggle switches to a 10,000-cell canvas-rendered density heatmap with rollups and a ranked exceptions list — aggregation with drill-down, not enumeration.

6.3 The six trigger scenarios

Simulate load spike (MLB)	PRB to 88%, congested. Real-world: a sudden demand surge. MLB offloads idle-mode users above its 75% threshold.
Simulate sleeping cell (ESM)	PRB to 8%. Real-world: a mostly-idle tower overnight. ESM applies micro cell-sleep below its 20% threshold — the industry's most quantifiable rApp value case.
Simulate cell outage (COC)	State to alarm, PRB to 95%. Real-world: a hard site failure. COC widens neighboring coverage via tilt while the fault is raised for physical repair.
Simulate handover failures (MRO)	HO failure rate to 5.5%. Real-world: drifted neighbor relations/parameters. MRO increases hysteresis above its 3.0% threshold until it settles.
Simulate missing neighbor (ANR)	Flags a missing cross-EMS neighbor relation. ANR discovers and adds it.
Simulate PCI collision (PCI)	Flags a PCI collision between nearby cells. PCI Reuse Optimization reassigns respecting minimum reuse distance.
Reset to nominal	Regenerates all 12 cells to healthy baseline — a neutral, demo-usability action.

6.4 Live conflict detection

Every simulation cycle, the platform checks whether more than one enabled rApp genuinely wants to act on the same cell at once. When it happens, it's a real detected conflict — logged live to the Conflict Log tab (Section 11) and resolved with whichever strategy is currently selected there.

6.5 Copilot narration

The R1 event feed toggles between a raw technical log and a plain-language narration of the same events.

7. Near-RT RIC Tab

Closes a gap: shows the fast half of RAN intelligence and the actual timescale contrast with the Non-RT RIC.

7.1 The explicit contrast

Closed Loop = Non-RT RIC, >1s timescale. This tab = Near-RT RIC, 10ms–1s over E2, steered only by slower A1 policy from above.

7.2 Making the timescale difference visible

A live "E2 Control Actions/sec" counter ticks every 150ms, versus the Non-RT RIC's roughly one action every few seconds.

7.3 The xApp catalog

Dynamic Spectrum Sharing	Real-time arbitration of shared spectrum between 4G and 5G carriers, slot by slot
Real-Time Load Balancing	Millisecond-scale traffic steering based on instantaneous radio conditions
Interference Mitigation	Detects and reacts to interference within the same control interval
QoS-Aware Scheduling Assist	Adjusts scheduler weighting in real time to protect latency-sensitive flows

8. rApp Catalog Tab

All six rApps have genuine, runnable reference logic.

rApp	Category	Live logic
Energy Saving Management	AI/ML forecast	PRB <20% sustained → micro cell-sleep mode
Automatic Neighbor Relation	Config Mgmt	Discovers/adds a flagged missing neighbor relation
Mobility Robustness Opt.	Perf Mgmt	HO failure >3.0% → +1dB hysteresis
Mobility Load Balancing	Perf Mgmt	PRB >75% → offload ~10% idle-mode users
Cell Outage Compensation	Fault/Self-Healing	Alarm state → neighbor tilt widened
PCI Reuse Optimization	Config Mgmt	PCI collision → reassign respecting reuse distance

All six are enabled by default; each card shows which R1-style data services it subscribes to.

9. Import rApp Tab

Demonstrates the "bring your own rApp" capability central to O-RAN's open ecosystem.

9.1 The descriptor schema

rAppId, name, version, requiredServices, exposedServices, policyType, and a logic object naming an algorithm and its parameters.

9.2 Honest handling of unrecognized logic

A recognized algorithm (one of the six built-ins) runs genuine logic; anything else is onboarded as an honest monitoring-only stub.

10. Slice Lifecycle Tab

1. Create	NSSI creation accepted, S-NSSAI allocated, RAN resource template applied
2. Modify	NSSI modification applied without service disruption
3. Monitor & Assure	Continuous SLA monitoring against committed targets
4. Decommission	NSSI decommission executed, resources released, inventory reconciled

10.1 Where each stage's inputs come from

A dedicated panel documents exactly what input each stage needs and where it comes from in a real deployment (business intent panel or NSMF/OSS order for Create; manual request or rApp recommendation for Modify; the same PM feed as FCAPS Monitor for Monitor & Assure; OSS/BSS order or automatic expiry for Decommission).

10.2 Business intent translation

Three example business intents translate into concrete 5QI/PDB/scheduling-weight policy, with a live bar chart against each active slice's committed SLA.

11. Conflict Log Tab

11.1 Two sources of conflicts

A manual "simulate a conflict now" button (ESM vs. MLB), plus live detection: the Closed Loop tab checks every cycle whether two enabled rApps genuinely want to act on the same cell, logging it here automatically.

Allow	All conflicting configurations applied; SMO monitors resulting KPI drift
First-Come-First-Served	The first-applied action is retained; others rejected for a cooldown window
Priority-Based Override	A pre-configured priority ranking decides the winner regardless of timing

11.3 Priority order for real, detected conflicts

Highest first: Cell Outage Compensation, Mobility Robustness Optimization, Mobility Load Balancing, Energy Saving Management, Automatic Neighbor Relation, PCI Reuse Optimization.

12. Data & Openness Tab

A data coverage matrix (per domain, per vendor: exposed vs. retained); a resilience/decoupling outage simulator feeding the Openness Scorecard's "Decoupled architecture" score; and self-serve data routing with no simulated ticket.

13. Cloud & NF Lifecycle Tab

A select-an-object model over a live, scrollable 100-object inventory across seven simulated locations. Instantiate always creates new; Heal/Scale/Upgrade/Terminate require a selection. Status self-drifts — a running object can spontaneously degrade, and transient states settle back to running after ~9s if untouched.

13.1 Where an Instantiate request's data comes from

Software package/signature from the container registry; hardware auto-discovered over O1 Plug-and-Play; configuration from planning-tool inputs via one templating engine; KPI validation reads the same PM feed as FCAPS Monitor. In practice triggered by a Zero-Touch Rollout run, a Slice Lifecycle capacity change, or a manual click here.

14. Zero-Touch Rollout Tab

A 13-step governed commissioning pipeline, from onboarding the software package through auto-discovery, deployment, activation, verification, KPI validation, and a saved audit baseline — sharing the same data-provenance chain as Cloud & NF Lifecycle.

15. Security & Trust Tab

Zero-trust access (NIST SP 800-207)	Every service call authenticated/authorized — zero implicit-trust paths
Component hardening (CIS)	Images scanned against a CIS baseline, findings remediated
Supply-chain integrity	SBOM generated, signatures verified, zero unverified artifacts
Secrets management	Encrypted at rest/in transit, access logged, rotated on schedule

Design principle: evidence, not assertion. A supply-chain artifact-verification log cycles a round-robin deployment queue, with a "next in queue" indicator so selection isn't a mystery.

16. Openness Scorecard Tab

Openness & data ownership	% of the Data & Openness coverage matrix marked exposed
Decoupled architecture	Rises with each successful reconnect after a simulated outage
Trustworthy automation	% of active rApps running genuine (non-stub) logic
Security by evidence	% of Security & Trust controls currently showing evidence
Safe autonomy	Strength of the active conflict-resolution strategy
Freedom from lock-in	% of Overview clusters that are actually interactive vs. documentation-only

Dragging a slider overrides that dimension manually until re-dragged or "Recalculate from live demo state" is pressed, which discards all overrides.

17. Documentation Tab

The in-app counterpart to this document: Glossary, Tab-by-tab guide, Scenario reasoning, Design principles, and a two-column Acronym index, alphabetized as one continuous sequence (left column, then right).

18. 50 Design Questions Tab

224 sub-questions across 50 topics in 8 themes, grounded in O-RAN/3GPP/TM Forum/ETSI/NIST/CIS material, each linked to exactly where this demo answers it — including honest "partial" or "out of scope" answers where warranted.

19. Design Principles

One file, no backend	Every tab, simulation, and calculation runs client-side in a single HTML file. Nothing can fail due to network issues, API limits, or a backend outage.
Honesty about synthetic data	Every screen is explicitly labeled simulation; unrecognized imported logic is labeled an honest stub, never silently faked.
Grounded in public standards only	O-RAN Alliance, 3GPP, TM Forum, NIST, CIS — no proprietary vendor material or trademarked product names.
Scale reasoning: aggregation over enumeration	A density heatmap, rollup stats, and ranked exceptions — not 10,000 individual dots.
Executive readability alongside technical depth	Plain-language framing throughout, without diluting the technical depth available on request.
Visible confirmation of every action	Button animation, toast notification, and (for Closed Loop) a shockwave highlight on the affected element.
Real detection, not scripted outcomes	Conflicts and faults are logged because the underlying simulation state actually produced one.
State that resolves on its own, imperfectly	Infrastructure can spontaneously degrade and transient states settle back to normal — closer to real infrastructure behavior.
A fixed, predictable layout footprint	Scrollable panels are sized consistently so the page doesn't shift as data accumulates.

19.10 Domain parity, not just RAN depth

Extending the pattern to Core and Transport meant holding every domain to the same bar rather than treating RAN as the "real" domain and the others as an afterthought: equal alarm/config-change counts across all three in FCAPS Monitor (Section 20.9), a rebalanced glossary and acronym index (Sections 2.11–2.13), and Core/Transport sections in Scenario Reasoning that match RAN's in depth, not just in existence.

19.11 Isolation over integration, where collision risk is real

The Cross-Domain concept is embedded as a sandboxed iframe rather than merged into the main app's DOM, specifically because both experiences reuse many of the same element IDs and JavaScript globals. Isolating them was judged safer and more maintainable than renaming everything to merge two large, independently-evolving codebases into one.

20. The Cross-Domain SMO Concept

Everything above (Sections 3–19) is the original, RAN-only Korognai SMO site. This section documents the second experience shipped alongside it: a forward-looking concept build that takes the identical SMO/Non-RT RIC closed-loop philosophy and extends it, for the first time, across all three domains that make up a real mobile network — RAN, Core, and Transport — under one control plane.

*Framing, stated up front: this is explicitly a **concept preview**, not a claim that any of this is deployed, fully standardized, or industry-consensus. Several of its names — covered in full in 20.1.1 below — are this build's own extension of the concept, not O-RAN Alliance standard terminology.*

20.1 How to reach it, why an iframe, and a note on naming

A pulsing card beside the main site's header — "Want to see into the future? Check the Cross-Domain 'SMO' concept" — opens a full-screen overlay containing the entire Cross-Domain build, loaded as an isolated document rather than merged into the RAN app's own page.

This is a deliberate technical choice, not an oversight: the two experiences independently reuse many of the same element IDs and JavaScript variable names (alarm lists, conflict logs, the rApp catalog object, and so on). Merging them into one DOM would risk one experience's script silently overwriting or reading the other's state. Loading the Cross-Domain build inside a sandboxed `<iframe>` keeps both fully independent while still shipping as a single self-contained HTML file — the cross-domain content is embedded as a base64-encoded payload and decoded into the iframe only when the card is clicked.

The Cross-Domain header itself reads, on two lines: "Cross-Domain SMO: RAN, Transport & Core — / Assurance, Automation & Intelligence, Built for What's Next."

20.1.1 A note on naming — what's real, what's this build's own shorthand

"Non-RT CIC" is not an O-RAN term — it's this build's own label. O-RAN Alliance's actual, standardized name for this layer is the **Non-RT RIC** (Non-Real-Time RAN Intelligent Controller). This build renames it "CIC" (Cross-Domain Intelligent Controller) purely as a naming device, because the whole point of this concept is that the same layer coordinates apps across RAN, Core, *and* Transport, not just radio — "RIC" literally stands for *RAN* Intelligent Controller, which stops quite fitting once it's managing cApps and tApps too. Nothing about the underlying architecture changes; it is the identical Non-RT RIC, relabeled for this demo's cross-domain framing.

In the same spirit, **rApps are the one genuinely standardized app family here** — defined by O-RAN Alliance for the RAN domain. This build extends that pattern to Core (**cApps**) and Transport (**tApps**), but those two names are this build's own invention, not yet an industry-agreed standard the way rApp/xApp are. They follow the identical detect → decide → act logic deliberately, so the pattern generalizes cleanly — but a reader checking this against real O-RAN documentation won't find "cApp" or "tApp" as terms there yet.

*One name in this document genuinely is the real, unaltered O-RAN term: **Non-RT RIC Federation** (Section 20.2.1, 20.3.5) — O-RAN Alliance's actual, emerging WG2 work on letting multiple Non-RT RICs, across vendors or regions, share intelligence instead of operating as isolated silos. It is called out by its real name specifically because it is an actual industry reference point, not this build's own naming convention — the two should not be confused with each other despite the similar-looking names.*

20.1.2 The Cross-Domain Overview tab itself

The first tab a visitor lands on inside the overlay. Fifteen functional-cluster cards cover every tab in both domains and every cross-domain tab — RAN/Core/Transport Architecture, Cross-Domain Topology, FCAPS & Assurance, Cloud & Automation, rApps/cApps/tApps Closed-Loop, Near-RT RIC, Slice Lifecycle, Intent-Based Assurance, Cross-Domain Conflict, Security & Trust, Openness Scorecard, Future & Standards Radar, and Documentation — each one clickable, jumping straight to that tab. This gives RAN, Core, Transport, and the cross-domain material equal visual weight, rather than making RAN look like the only fully-featured domain.

The naming-honesty note from Section 20.1.1 above is also surfaced here, in-app, on this same tab — not just in this document — since it's exactly the point where a first-time visitor is deciding how much weight to put on the

"CIC"/"cApp"/"tApp" terminology they're about to see throughout the rest of the concept.

20.2 Cross-Domain Topology Tab

Answers the RAN Architecture tab's question one level up: where does one SMO sit when it spans three domains, and what ties it together? A single diagram shows SMO / Non-RT CIC (this build's own name for the layer — see 20.1.1 — plus the real Non-RT RIC Federation function) presiding over three domain blocks — RAN, Transport, Core — each with its own local controller/analytics function, plus a per-domain availability/alarm strip linking straight through to that domain's own tab.

20.2.1 Cross-domain interface reference

Interfaces already covered by the RAN interface table (O1/O2/A1/E2) are not repeated here — only the interfaces this cross-domain footprint adds beyond those:

Interface	Connects	What it's for
TAPI	T-SDN ↔ SMO	Transport API (ONF) — requests/manages optical/packet transport paths; the transport equivalent of O1/O2
TS 28.312 intent	Operator/automation ↔ SMO	3GPP SA5 intent-driven management — state a goal, SMO translates it into per-domain configuration
Nnwdaf	NWDAF ↔ cApps	Analytics exposure — the interface cApps ultimately consume core predictions through
Non-RT RIC Federation	Non-RT RIC ↔ Non-RT RIC	Lets multiple Non-RT RICs (across vendors/regions) share rApp intelligence rather than operating as silos
CAMARA / Open Gateway	3rd-party apps ↔ SMO	GSMA/Linux Foundation developer APIs exposing network capability, fulfilled by rApps/cApps/tApps under the hood
SEPP	Core ↔ roaming partner	Trust boundary for inter-operator signaling — the core-domain analogue of a RAN security gateway

20.3 Cross-Domain Loop Tab

The Cross-Domain build's centerpiece, and structurally the most significant change from the RAN-only site: it absorbs the entire RAN Closed Loop tab (the cell ring, fleet view, and all 6 rApp triggers — see Section 6) as its first section, then adds two more domain-local closed loops and a worked cross-domain scenario beneath it.

20.3.1 RAN section

Identical to Section 6 in full — the same cell ring, fleet toggle, six trigger buttons, and live rApp-vs-rApp conflict detection, presented here as one section of a larger tab rather than its own tab.

20.3.2 Core section — five cApp triggers

Simulate signaling storm	AMF-02 registration/session burst (mass-reconnect pattern) → Signaling Storm Mitigation cApp applies AMF-level throttling/back-off
Simulate UPF overload	UPF-05 throughput >92% → UPF Load Balancing cApp steers new sessions to a less-loaded UPF
Simulate QoS risk	NWDAF predicts a URLLC slice will breach its packet delay budget within 90s → NWDAF-Driven QoS Prediction cApp pre-emptively adjusts 5QI/GBR
Simulate slice over-admission	NSSF headroom for an enterprise slice drops below 10% → Slice Admission Control cApp tightens the threshold
Simulate subscriber anomaly	UDM flags an impossible-travel roaming pattern → Subscriber Anomaly & Fraud Detection cApp flags it for SOC review

20.3.3 Transport section — five tApp triggers

Simulate fronthaul congestion	FH-LINK-014 latency 118µs, above the 100µs O-DU split budget → Fronthaul Latency & Jitter Guard tApp reroutes via TAPI/PCE
Simulate spectrum fragmentation	DWDM-CH-22 fragmented across 3 blocks → Optical Spectrum Defragmentation tApp recomputes channel assignments
Simulate timing holdover	PTP-GM-02 grandmaster enters holdover → Timing/Sync Assurance tApp fails over to a backup source
Simulate slice path congestion	A slice's transport path sustained >90% of reserved bandwidth → Transport Slice Path Computation tApp reserves additional bandwidth
Simulate encryption fallback	A link falls back to cleartext after an IPsec renegotiation failure → IPsec/MACsec Assurance tApp flags it

All five cApps and all five tApps are enabled by default (matching the RAN app's six-rApps-enabled convention), so every trigger button resolves automatically; disabling an app on its own catalog first demonstrates the "condition detected, nothing subscribed" case instead.

Core and Transport each have their own ring diagram alongside their trigger buttons, matching the RAN cell ring's visual language (8 labeled nodes around a "Non-RT CIC" hub, live KPI ticker beneath). Nodes also drift into an alarm/congested state on their own every 9–11 seconds, roughly a fifth of the time — the same ambient behavior the RAN cell ring already had — but only when that domain's ring is currently all-nominal, so ambient drift resolves cleanly via the matching cApp/tApp rather than colliding with another condition already in progress on the same node.

20.3.4 Technical / Plain-language (Copilot) toggle, all three domains

Every section — RAN, Core, and Transport — has its own Technical/Plain-language log toggle, matching the RAN app's R1 event feed pattern (Section 6.5), so Copilot-style narration is available for cApp and tApp actions exactly as it is for rApps.

20.3.5 Cross-domain worked scenario

Beneath the three domain-local sections sits a scripted, worked example: an enterprise URLLC slice starts missing its latency SLA. Root cause could be radio congestion, a congested fronthaul link, or a slow UPF. A cross-domain correlation function (this build's Non-RT CIC, backed by NWDAF core analytics — not to be confused with the real Non-RT RIC Federation function, see 20.1.1) correlates PM data from all three domains within the same time window, produces a confidence-scored root cause (e.g. "78% fronthaul congestion, 15% RAN, 7% core"), and dispatches the fix to whichever domain actually owns it — a tApp reroute, an rApp offload, or a cApp scale-out — rather than three uncorrelated tickets landing on three different queues.

20.4 Cross-Domain Conflict Tab

Absorbs the RAN Conflict Log tab (Section 11) as its first section, adds matching Core and Transport sections, and a fourth section covering conflict *between* domains, a different kind of conflict from the same-domain cases above.

Section	Conflict type	What it models
RAN — rApp vs. rApp	Same object	Identical to Section 11: two rApps target the same RAN control parameter. Same three strategies (Allow / FCFS / Priority).
Core — cApp vs. cApp	Same object	Two cApps propose conflicting changes to the same core network function within a configurable window.
Transport — tApp vs. tApp	Same object	Two tApps propose conflicting path/spectrum changes on the same transport link.
Cross-domain	Different objects, shared outcome	One domain's fix undermines another domain's fix with *no shared object at all* — a class of conflict invisible to any single domain's own monitoring.

20.4.1 The three cross-domain scenarios

RAN ↔ Transport	Mobility Load Balancing reroutes traffic onto the exact fronthaul link a tApp is mid-defragmentation on, momentarily doubling its load
Core ↔ Transport	UPF Load Balancing steers sessions onto a UPF whose transport path a tApp had just reserved exclusively for a different slice
RAN ↔ Core	Cell Outage Compensation's capacity boost drives a registration/handover burst that Signaling Storm Mitigation reads as an attack and starts throttling — right when capacity is needed most

Each still resolves under the identical three strategies; Priority-Based Override here ranks one *domain's* assurance need above the other's, rather than one app's priority above another's within a single domain.

20.4.2 Two-way propagation into the Cross-Domain Loop tab

A conflict simulated here also logs a line into the relevant domain's Cross-Domain Loop section (Section 20.3) — e.g. a Core conflict appears in both the Cross-Domain Conflict tab's Core log and the Cross-Domain Loop tab's Core closed-loop log — so the same event is visible from either tab rather than siloed in just one.

20.5 Intent-Based Assurance Tab

A top-level tab implementing 3GPP TS 28.312 and TM Forum IG1230/TMF921's intent model: Intent → Expectations → Targets. Rather than the operator hand-writing settings for three separate domains, they state the outcome they want in plain language and let SMO work out what each domain needs to do to deliver it.

Real-world framing: in production, the "intent" input would be free text, parsed by an NLP/LLM layer into the same Expectations/Targets structure shown here. This demo uses five pre-built example intents (buttons, not a text box) so the resulting decomposition and live compliance behavior can be inspected deterministically — the underlying model is the same either way.

20.5.1 The five example intents

Intent	Category	Per-domain Targets
Guarantee low latency — factory robotics	URLLC	RAN ≤2.0% HO fail, Core ≤70% UPF, Transport ≤90µs FH
Guarantee throughput — premium video tier	eMBB	RAN ≤4.0%, Core ≤85%, Transport ≤110µs

Intent	Category	Per-domain Targets
Maximize capacity — stadium event	Capacity event	RAN ≤5.0%, Core ≤92%, Transport ≤125µs
Guarantee reliability — emergency services	Public safety	RAN ≤1.5%, Core ≤75%, Transport ≤85µs (tightest of the five)
Minimize energy — off-peak hours	Energy saving	RAN ≤6.0%, Core ≤90%, Transport ≤130µs (deliberately loosest — the goal here is power savings, not performance)

20.5.2 Decomposition and deployment

Selecting an intent renders its decomposition as a tree — Intent at the top, its Expectations beneath, each Expectation's concrete per-domain Target beneath that. A "Deploy this intent" button logs the decomposition into all three domains' Cross-Domain Loop logs simultaneously, the same propagation pattern used elsewhere (Section 20.3.4, 20.4.2).

20.5.3 Live assurance and genuine dispatch

Three KPI cards (RAN, Core, Transport) update every 1.5 seconds against the deployed intent's Targets, alongside a single Compliant / At Risk / Breached badge. This is not cosmetic: a breach calls the same closed-loop remediation logic the Cross-Domain Loop tab uses. If the responsible app (an rApp, cApp, or tApp) is enabled, it genuinely reduces the offending metric and logs the dispatch with an "Intent-Based Assurance breach →" attribution; if that app is disabled, the log honestly states the domain **will not self-correct** rather than silently pretending the breach resolved. A guard prevents the same breach from re-dispatching repeatedly while it persists, and resets the moment a new intent is selected or the domain recovers.

20.5.4 Competing intents

A short explanatory section plus a "simulate" button models the harder real-world problem this tab doesn't fully solve: two valid intents (e.g. maximize capacity vs. minimize energy) can imply opposite actions on the same domain at once. The simulated log illustrates the tension rather than resolving it outright — a deliberate acknowledgment that intent conflict is a genuinely open problem, not a solved one, in real intent-based networks too.

20.6 RAN & Apps / Core & Apps / Transport & Apps Dropdowns

The tab bar is reorganized around three domain dropdowns, each expanding to the identical five-tab pattern, so the navigation itself communicates domain parity rather than RAN having eleven tabs and the others having none.

Dropdown	Contains	Documented in
RAN & Apps	RAN Architecture, Near-RT RIC, rApp Catalog, Import rApp, Slice Lifecycle, Data & Openness, Cloud & NF Lifecycle, Zero-Touch Rollout, 50 Design Questions	Sections 4, 7–10, 12–14, 18
Core & Apps	Core Architecture, App Catalog, Import App, Data & Openness, Cloud & NF Lifecycle	Section 20.7
Transport & Apps	Transport Architecture, App Catalog, Import App, Data & Openness, Cloud & NF Lifecycle	Section 20.8

Conflict Log for each domain lives in the single top-level Cross-Domain Conflict tab (Section 20.4) rather than as separate tabs inside each domain's dropdown, so all three domains' conflict behavior is visible side by side.

20.7 Core & Apps — Architecture, Catalog, Import, Openness, Cloud

20.7.1 Core Architecture

A 5G Core Service-Based Architecture diagram (NRF, AMF, SMF, UPF, PCF, UDM, NSSF, NWDAF on a common service bus), plus two reference tables — core network functions and their roles, and the core interfaces (N1–N4, N11, N15, Nnwdaf, N32) — mirroring RAN Architecture's diagram-plus-interface-table pattern exactly. Fault and performance detail for the Core domain lives in FCAPS Monitor (Section 20.9) instead, alongside RAN's and Transport's, rather than being duplicated here.

20.7.2 App Catalog — 5 cApps

cApp	Category	Reference algorithm ID
UPF Load Balancing cApp	Traffic Engineering	upf-load-balancing
Signaling Storm Mitigation cApp	Resilience	signaling-storm-mitigation
Slice Admission Control cApp	Slicing	slice-admission-control
NWDAF-Driven QoS Prediction cApp	AI/ML	nwdaf-qos-prediction
Subscriber Anomaly & Fraud Detection cApp	Security	subscriber-anomaly-detection

20.7.3 Import App

Identical validation pattern to Import rApp (Section 9), generalized to the core domain: appId, name, version, requiredServices, exposedServices, policyType, logic.algorithm/params. The textarea is pre-loaded with a working sample descriptor by default. A recognized algorithm (the five above) runs genuine logic; anything else onboards as an honest stub.

20.7.4 Data & Openness

A core telemetry coverage matrix (open via NWDAF/SBA vs. retained, e.g. subscriber profile data in UDM) plus an NWDAF analytics-feed disconnect/reconnect simulator, mirroring the RAN Data & Openness tab's pattern (Section 12).

20.7.5 Cloud & NF Lifecycle

A smaller-scale sibling of the RAN Cloud & NF Lifecycle table (Section 13): a live table of AMF/SMF/UPF/PCF/UDM/NRF/NSSF/NWDAF instances with Instantiate/Heal/Scale/Upgrade/Terminate actions and a selected-object banner.

20.8 Transport & Apps — Architecture, Catalog, Import, Openness, Cloud

20.8.1 Transport Architecture

A fronthaul/midhaul/backhaul plus T-SDN controller diagram, plus a transport network elements reference table (T-SDN controller, O-RU, O-DU/O-CU aggregation, DWDM node, PTP grandmaster, backhaul router) and an interface reference table (TAPI, fronthaul/midhaul/backhaul, PTP/SyncE) — the same restructuring rationale as Core Architecture (Section 20.7.1) applies here.

20.8.2 App Catalog — 5 tApps

tApp	Category	Reference algorithm ID
Fronthaul Latency & Jitter Guard tApp	Assurance	fronthaul-latency-guard
Optical Spectrum Defragmentation tApp	Optimization	spectrum-defragmentation
Transport Slice Path Computation tApp	Slicing	slice-path-computation
Timing / Sync Assurance tApp	Resilience	timing-sync-assurance
IPsec/MACsec Assurance tApp	Security	ipsec-macsec-assurance

20.8.3 Import App

Same schema and genuine-vs-stub logic as Core's Import App (Section 20.7.3), scoped to the five tApp algorithms above; also pre-loaded with a working sample descriptor by default.

20.8.4 Data & Openness

A TAPI telemetry coverage matrix plus a transport analytics-feed disconnect/reconnect simulator.

20.8.5 Cloud & NF Lifecycle

A live table of T-SDN controller, DWDM nodes, PTP grandmasters, aggregation switches, and backhaul routers, with the identical Instantiate/Heal/Scale/Upgrade/Terminate action set.

20.9 FCAPS Monitor (Cross-Domain)

Opens with a cross-domain FCAPS summary table rolling up Fault/Configuration/Performance/Security posture for all three domains at a glance (status only — no action buttons; the tab bar is how you get to detail). Below that, Fault, Performance, and Configuration Management are genuinely *unified* feeds, not three separate views: every RAN, Core, and Transport alarm, KPI, and config change sits in the same three lists, each row carrying a color-coded domain chip (amber RAN, cyan Transport, purple Core).

Feed	Composition	Note
Active alarms	20 RAN + 20 Core + 20 Transport = 60 total	Equal counts across domains, deliberately — no domain is treated as more important than another
Live KPIs	5 RAN + 5 Core + 5 Transport = 15 total	Equal counts across domains, for the same reason
Configuration changes	20 RAN + 20 Core + 20 Transport = 60 total	Same equal-count principle as alarms

Alarms render as a compact table (age, domain, severity, ID/source, description, actions) — matching Configuration Management's row style — for a scannable, dense view given that three domains share one list.

Trace sessions and topology/inventory remain RAN-scoped by design, since a trace follows one device's radio session and Core/Transport each have their own Architecture-tab reference material instead.

20.10 Security & Trust (Cross-Domain)

The same evidence-based four controls as the RAN app (Section 15), plus a fifth: **Cross-domain interface security** — O-RAN WG11 controls on E2/O1/A1, 3GPP's SEPP enforcing N32 roaming security, and TLS 1.3-only TAPI sessions for transport control.

The supply-chain artifact-verification queue holds twelve artifacts total: four RAN examples plus Core (cApp package, 5GC/NWDAF Helm chart, a core container image, a core SBOM manifest) and Transport (tApp package, T-SDN controller Helm chart, a transport container image, a transport SBOM manifest), all in the same round-robin queue.

20.11 Openness Scorecard (Cross-Domain)

Identical six principles and mechanics to Section 16, with one difference: **Trustworthy automation** is computed across every enabled rApp, cApp, and tApp together — genuine-vs-stub logic is checked against all three domains' reference algorithm lists, not just RAN's — so enabling or disabling a cApp or tApp moves this score exactly the way toggling an rApp does.

20.12 Future & Standards Radar Tab

Has no equivalent elsewhere in either experience. Two parts:

20.12.1 Autonomy ladder self-placement

An honest L0–L5 self-assessment against TM Forum's Autonomous Networks maturity model (the same shape as the well-known SAE self-driving-car scale, applied to network operations). This build places itself at **L4, high autonomy**: per-domain closed loops with conflict management, plus a genuine cross-domain correlation engine (Section 20.3.5) that manages a scenario end to end without a human in the loop — reserving L5 (full autonomy, zero exceptions, every domain and timescale) as an honest ceiling this build does not claim.

20.12.2 Forward-looking radar

- AI-native RAN / AI-native air interface (3GPP AI/ML air-interface work, the AI-RAN Alliance industry body)
- 6G framework and early standardization (ITU-R IMT-2030, 3GPP's roadmap toward it)
- Network Digital Twin (ITU-T and operator-led initiatives, not yet in this build)
- Non-RT RIC Federation (O-RAN's emerging multi-vendor/multi-region federation work)
- Intent-based, closed-loop management (TM Forum IG1230/TMF921, 3GPP TS 28.312, ETSI ZSM converging on the same idea) — demonstrated as a working preview on the Intent-Based Assurance tab (Section 20.5), not just referenced as a standard
- Network APIs for developers (GSMA Open Gateway / CAMARA)
- Energy & sustainability standardization (ITU-T L.1381-style metrics, operator net-zero commitments)
- Post-quantum-ready O-RAN security (O-RAN WG11 scoping PQC-resilient key exchange)

A closing table lists the standards bodies (O-RAN Alliance, 3GPP SA5, TM Forum, ETSI ZSM, ONF, GSMA, ITU-R/ITU-T) referenced across the whole cross-domain expansion — see Section 21 for the consolidated version.

20.13 Documentation Tab (Cross-Domain)

The in-app documentation for this experience matches the structure of this printed document:

- **Glossary** — covers Core, Transport, Cross-domain/standards, and Intent-based management terms alongside RAN's (matching Sections 2.11–2.14 above, plus the naming note in 20.1.1), rather than treating RAN as the only documented domain.
- **Tab-by-tab guide** — a row per tab, including every Core & Apps / Transport & Apps dropdown entry individually rather than one combined row per domain, plus a row for Intent-Based Assurance.
- **Scenario reasoning** — RAN (6), Core (5), and Transport (5) trigger-button write-ups, matched in depth, plus a write-up for each of the five example intents.
- **Acronym index** — 71 terms (AMF, SMF, UPF, NWDAF, SBA, TAPI, DWDM, PTP, ETSI, GSMA, TM Forum, PQC, CIC, and more), alphabetized as one continuous sequence, left column then right (Section 21.4 has the full list).
- The header's "New to all this?" entry point is present on the RAN app only.

20.14 Deployment, Portability & Mobile

Both experiences are designed to be moved between hosts freely, since the whole point of shipping as a single static HTML file is that it should run the same way anywhere.

20.14.1 URL hash deep-linking

Every tab in both experiences can be linked to directly via a URL hash, e.g. `yourdomain.com/#intent-based-assurance` or `yourdomain.com/#core-apps`. Since the Cross-Domain concept runs inside an isolated `<iframe>` (Section 20.1), it can't rewrite the parent page's own URL directly — the parent opens the overlay and tells the iframe which tab to show via `postMessage`, and the iframe reports back the same way whenever the visitor clicks around inside it, so the address bar stays in sync in both directions.

20.14.2 Host and domain independence

Hash fragments are never sent to the server, so this works with zero server-side configuration on GitHub Pages, any other static host, or a custom domain via CNAME — and switching hosting providers or domains entirely requires no code changes, since nothing in either file references a specific URL.

20.14.3 Mobile responsiveness

Both experiences are responsive down to phone width: the header stacks into a single column instead of overlapping, the tab bar becomes a horizontally scrollable strip instead of wrapping awkwardly, and hover-only tooltips are suppressed on touch devices instead of getting stuck open after a tap.

21. Appendix: Full Interface, App & Acronym Reference Tables

21.1 RAN interface quick reference

Interface	Connects	Timescale / notes
O-FH	RU ↔ DU	Fronthaul, microsecond-scale, functional split 7.2x
F1	DU ↔ CU	Midhaul, single-digit ms
E1	CU-CP ↔ CU-UP	Internal CU split
E2	Near-RT RIC ↔ RAN	10ms–1s, xApp control channel
A1	Non-RT RIC ↔ Near-RT RIC	Policy guidance, >1s
O1	SMO ↔ RAN elements	FCAPS
O2	SMO ↔ O-Cloud	Infrastructure/workload lifecycle
R1	rApps ↔ Non-RT RIC	Data/service exposure (SME/DME) — still maturing in O-RAN standards

21.2 Core & Transport interface quick reference

Interface	Connects	Notes
N2 / N3 / N4	RAN↔AMF / RAN↔UPF / SMF↔UPF	Core reference points; N4 carries PFCP
N32	SEPP ↔ SEPP	Inter-operator roaming security edge
Nnwdaf	NWDAF ↔ cApps	Core analytics exposure
TAPI	T-SDN ↔ SMO	Transport path/topology management (ONF)
PTP / SyncE	Across the transport network	Timing distribution

21.3 App quick reference (all 16, by domain)

App	Domain	Trigger condition in this demo
Energy Saving Management (ESM)	RAN	Sustained PRB utilization below 20%
Automatic Neighbor Relation (ANR)	RAN	A cell flagged as missing a neighbor relation
Mobility Robustness Optimization (MRO)	RAN	Handover failure rate above 3.0%
Mobility Load Balancing (MLB)	RAN	PRB utilization above 75%
Cell Outage Compensation (COC)	RAN	A cell in alarm state
PCI Reuse Optimization (PCI)	RAN	A cell flagged with a PCI collision
UPF Load Balancing cApp	Core	UPF throughput sustained above 92%
Signaling Storm Mitigation cApp	Core	Registration/session-request burst
Slice Admission Control cApp	Core	NSSF headroom drops below 10%

App	Domain	Trigger condition in this demo
NWDAF-Driven QoS Prediction cApp	Core	Predicted packet-delay-budget breach
Subscriber Anomaly & Fraud Detection cApp	Core	Impossible-travel roaming pattern
Fronthaul Latency & Jitter Guard tApp	Transport	Fronthaul latency above the O-DU split budget
Optical Spectrum Defragmentation tApp	Transport	Spectrum fragmented across non-contiguous blocks
Transport Slice Path Computation tApp	Transport	Path sustained above 90% of reserved bandwidth
Timing / Sync Assurance tApp	Transport	Grandmaster clock enters holdover
IPsec/MACsec Assurance tApp	Transport	A link falls back to cleartext

21.4 Full acronym index (alphabetical, 71 terms)

Matches the index shown in-app — alphabetized as one continuous sequence, left column then right.

5QI	5G QoS Identifier	**OSNR**	Optical Signal-to-Noise Ratio
AMF	Access & Mobility Mgmt Function	**OSS**	Operations Support System
ANR	Automatic Neighbor Relation	**PCF**	Policy Control Function
BSS	Business Support System	**PCI**	Physical Cell Identity
CA	Carrier Aggregation	**PDB**	Packet Delay Budget
CIC	Cross-Domain Intelligent Controller (this build's name for Non-RT RIC)	**PER**	Packet Error Rate
CIS	Center for Internet Security	**PFCP**	Packet Forwarding Control Protocol
CNF	Cloud-native Network Function	**PnP**	Plug-and-Play
COC	Cell Outage Compensation	**PQC**	Post-Quantum Cryptography
DL / UL	Downlink / Uplink	**PRB**	Physical Resource Block
DME	Data Management & Exposure	**PTP**	Precision Time Protocol
DWDM	Dense Wavelength Division Multiplexing	**RAN**	Radio Access Network
eMBB	Enhanced Mobile Broadband	**RF / PHY**	Radio Frequency / Physical Layer
ESM	Energy Saving Management	**RIC**	RAN Intelligent Controller
ETSI	European Telecom. Standards Institute	**RU/DU/CU**	Radio/Distributed/Centralized Unit
FCAPS	Fault, Config, Acct, Perf, Security	**SBA**	Service-Based Architecture
GBR	Guaranteed Bit Rate	**SBI**	Service-Based Interface
GSMA	GSM Association	**SBOM**	Software Bill of Materials
HO	Handover	**SEPP**	Security Edge Protection Proxy
IMT-2030	ITU-R's 6G framework	**SLA**	Service Level Agreement
ITU-R / ITU-T	Int'l Telecom Union sectors	**SME**	Service Management & Exposure
MLB	Mobility Load Balancing	**SMF**	Session Management Function
mMTC	Massive Machine-Type Comm.	**SMO**	Service Mgmt & Orchestration
MRO	Mobility Robustness Optimization	**S-NSSAI**	Slice Selection Assistance Info
N2/N3/N4	Core reference points	**SON**	Self-Organizing Network
N32	Roaming reference point (SEPP)	**SyncE**	Synchronous Ethernet
NF / LCM	Network Function / Lifecycle Mgmt	**TAPI**	Transport API
NIST	Nat'l Institute of Standards & Tech.	**TCO**	Total Cost of Ownership
NRF	Network Repository Function	**TM Forum**	TeleManagement Forum
NSSF	Network Slice Selection Function	**T-SDN**	Transport SDN
NSSI	Network Slice Subnet Instance	**UDM**	Unified Data Management
NWDAF	Network Data Analytics Function	**UE**	User Equipment

O2-DMS	O2 Deployment Mgmt Service	**UPF**	User Plane Function
O-FH	Open Fronthaul	**URLLC**	Ultra-Reliable Low-Latency Comm.
ONF	Open Networking Foundation	**ZSM**	Zero-touch Network & Service Mgmt
OPEX	Operating Expenses		